

Stat 371 Practice Problem Session

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Problem 0: Getting Started

Set up R / R studio if you haven't already. Now is a great time to get help if you're having trouble.

1. First run a line of code like:

```
In [ ]: print("Hello world")
```

2. R can be used like a calculator!

Try using it like one! The probability of rolling three consecutive 6's on a fair die is shown below.

```
In [ ]: (1/6)^3
```

3. Now we'll see some basic R commands. First we'll assign a list to a **variable**:
It looks like

```
variable_name <- c(list of numbers that's comma separated)
```

```
In [7]: # This saves the list of numbers to your variable
number_list <- c(1,2,3)

# This prints the variable
print(number_list)
```

```
[1] 1 2 3
```

Try for yourself. Save the following data into a variable called "list_A"

A : 10, 11, 12, 12, 13, 14, 15

4. Some of the problems that we'll see today will use Numerical Summaries like **Mean, Median, Variance**, and **standard deviation** (Variance/standard

deviation describe how 'spread out the data is'. You'll see them in class this week). You can compute these quantities like so:

```
In [25]: mean <- mean(number_list)
median <- median(number_list)
variance <- var(number_list)
standard_deviation <- sd(number_list)

# Here "paste()" combines the text and numbers into one big string of
textoutput <- paste("Mean: ", mean, " Median: ", median, " Variance: "
print(textoutput)
```

```
[1] "Mean:  2  Median:  2  Variance:  1  Standard Deviation:  1"
```

Try computing these quantities for the data in list_A!

5. Final Notes:

- Notice how the variable assingment is "<=". You can also use "=". Try replacing "<=" in one line of your code
- Notice how some of the code above starts with a hastag #. You can create comments by placing these on their own line or after your code *# I'm a comment.*

```
In [ ]: 1 + 1 # I am a comment and I'm invisible to the R interpreter. R doesn
```

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Problem 1: Graph First

The following hypothetical data are germination times, in days, for 12 seeds grown under the same laboratory condition:

5, 6, 6, 7, 7, 7, 8, 8, 9, 9, 10, 17.

The code below displays the datan in a boxplot. Try running it

```
In [ ]: germination_days <- c(5, 6, 6, 7, 7, 7, 8, 8, 9, 9, 10, 17)

stripchart(
  germination_days,
  method = "stack",
  pch = 19,
  xlab = "Germination time (days)",
  main = "Germination Times for 12 Seeds"
)
```

- (a) Describe the distribution's **shape** and identify any important unusual feature.
- (b) Before calculating anything, which do you expect to be larger: the **mean** or the **median**? Explain using the graph.
- (c) A student reports only the mean germination time and says that this completely describes the dataset. Explain why a graph adds information that the mean alone cannot show.

Problem 2: What Does One Extreme Value Change?

Consider the following two hypothetical datasets.

A : 10, 11, 12, 12, 13, 14, 15

B : 10, 11, 12, 12, 13, 14, 42

Dataset B is identical to Dataset A except that the largest value, 15, has been replaced by 42.

(a) Before doing any calculations, predict what will happen to each of the following when moving from Dataset A to Dataset B:

1. mean;
2. median;
3. standard deviation;

For each quantity, state whether you expect it to change substantially, change only a little, or remain unchanged.

(b) Write some R code to compute the mean and medians of these datasets

(c) Which two of these four statistics are **resistant** to the extreme value in this example? Explain what resistant means.

(d) Suppose Dataset B came from a real biological study. Should the value 42 automatically be deleted because it is extreme? Explain.

Problem 3: Viral Mutations

A researcher at the Cleveland Clinic is studying intra-host mutations of SARS-CoV-2. They recruit 589 COVID-19 patients from the greater Cleveland area and collect nasal swab samples at initial diagnosis and again 2 weeks later. Using

deep sequencing, the researcher counts the number of new mutations that were present in the viral Spike (S) gene after the 2-week period for each patient and reports their average across the sample.

Describe the following:

1. Population, sample, and unit of observation (or 'case')
2. Parameter and Statistic
3. Is this a simple random sample?
4. Is this an experimental or observational study?

Problem 4: Confounding Ballers

A confounding variable is a variable that may look like it's connected to a variable of interest, but isn't. I include an hypothetical scenario with a confounder:

A 2026-2027 NBA rule change requires teams to play a non-profesional player for five minutes a game. Miraciously you are chosen to play and travel with the San Antonio Spurs. You notice that the year of this change they have a 68-14 Win-Loss record at the end of the regular season, which is an improvement over the 62-20 W/L record they had last year. They don't have any other big changes to their team, so you conclude that you were the key to their improved performance even though you scored 0 points the entire season. **What is the confounding variable in this example?** Explain what might be wrong about this conclusion.

Bonus Problems

Problem 5: Confounding Scenarios

Not all confounding variables are so obvious as the example above. Think about possible confounders for the following scenarios

1. Taking a daily vitamin D supplement reduces the risk of cardiovascular disease
2. A specific single nucleotide polymorphism (SNP) causes increased height
3. You catch 10 fish of the same species from Lake Mendota. You move 5 to the Atlantic Ocean and 5 to Lake Superior. You observe that the fish moved to the Atlantic die soon after release and conclude that these fish are sensitive to being moved long distances.